PALADINMEDIA: Energy-Efficient Post-Quantum Media Steganography System

Project Status, Technical Architecture & Milestone Completion Audit Report

Current Baseline: MVP 0 / Phase 2 Verified	**Automated Pytest:** 99/99 Passed (100%)	**Sprint 1 Invariants:** 24/24 Passed (100%)	**Core Delivery:** 100% (Phases 0–2 Complete)

1. Executive Summary & Project Purpose

Overview: PaladinMedia is an advanced media concealment system designed for secure, reversible transport of classified digital content (text, JSON, high-entropy binary, and raster images) concealed inside benign-looking 512×512 RGB carrier images. Standard encryption exposes conspicuous high-entropy ciphertext to network censors, while traditional steganography lacks provable cryptographic confidentiality and noise tolerance. PaladinMedia solves both challenges simultaneously.

Core Architectural Principle — 'Security Before Concealment': The system enforces an invariant sequence: payloads are first canonically framed with wire headers and CRC-32 checksums, encrypted using authenticated AES-256-GCM with associated data (AAD), encoded with systematic Reed-Solomon GF(2^8) error correction, and only then embedded into the discrete wavelet transform subbands (2D Haar DWT) of a carrier image. The receiver executes a strictly symmetric extraction pipeline, guaranteeing bit-exact secret recovery.

Engineering Baseline & Governance: *The project strictly adheres to the principle 'Hypothesis ≠ Implementation ≠ Experiment ≠ Validation.' Development is governed by three frozen specifications:* ***Product Vision*** *(user scope and boundaries),* ***Architecture Specification*** *(module contracts and fail-closed interfaces), and* ***System & Research Document*** *(immutable evidence ledger for empirical metrics and reversibility validation).*

2. Current Implementation Status & Workflow

The software architecture is fully operational across three modular tiers behind frozen, stable interfaces:

Pipeline Stage	Engineered Module & Location	Implemented Functionality & Technical Guarantees
1. Wire Framing	SecretSerializer src/core/serialization	Packs secret media with canonical b'PLDS' magic header, version byte, media type enum, length, and CRC-32 trailer.
2. Cryptography	CryptoProvider src/security/crypto	AES-256-GCM cipher with 96-bit CSPRNG nonce and 128-bit GHASH authentication tag bound to transcript AAD. Fail-closed security.
3. Error Correction	ECCEncoder src/security/ecc	Systematic Reed-Solomon GF(2^8) chunking with 2t=32 parity symbols per 255-byte block. Repairs up to t=16 symbol errors (128 bit flips).
4. Wavelet Transform	HaarDWT2D src/carrier/dwt	1-level 2D Haar DWT in float64 precision decomposing 512×512 RGB carrier into LL, LH, HL, HH subbands (3×256×256) with zero inversion error.
5. Carrier Modulation	Embedder & Mask src/carrier	16-byte fixed canonical b'PLDH' header anchored in first 128 positions; deterministic PRNG position mask across HH subband.
6. Symmetric Receiver	MediaPipeline src/core/media_pipeline	Header extraction → mask reconstruction → subband extraction → RS error correction → GHASH tag validation → 100% exact byte recovery.

Capacity Accounting & Hard Gating: A 512×512 carrier provides 65,536 bits (8,192 B) of raw capacity in 1 HH subband channel. Deducting the 16 B carrier header establishes a hard capacity limit of 8,176 B for the protected payload (~7,088 B secret data). Payloads exceeding this threshold trigger a hard gate (PayloadCapacityExceededError) with zero carrier modifications.

3. Verification & Empirical Testing Results

System functionality has been validated across three independent test regimes: an automated 99-test Pytest suite, a 24-check cryptographic invariant verification script (verify_sprint1.py), and formal steganography experiment **E03**.

Verification Dimension	Metric & Standard	Measured Value	Baseline Standard	Engineering Finding & Verification Result
Automated Test Suite	Pytest Execution	**99 / 99 Passed (100%)**	100% Pass	Full coverage across unit, boundary, fail-closed, and roundtrip tests (2.38s).
Sprint 1 Invariants	**Security Assertions**	**24 / 24 Passed (100%)**	INV-01 .. INV-03	Verified fail-closed zero-plaintext behavior under malicious fault injection.
Transform Precision	Haar Inversion Error	**0.000000 max error**	Exact 0.000	Float64 matrix transform exhibits complete mathematical reversibility.
Perceptual Stealth	Carrier Distortion (PSNR)	**75.71 dB**	> 50.0 dB	Carrier distortion is strictly imperceptible; exceeds industry stealth threshold.
Structural Similarity	Image Geometry (SSIM)	**1.00000**	> 0.9990	Identical luminance and structural contrast preserved across all color channels.
Channel Noise	Pre-ECC BER	**0.000000**	< 0.01	Zero bit flips introduced across lossless PNG encoding, disk save, and reload.
Noise Resilience	Reed-Solomon Repair	**Repairs t=16 symbols**	Up to 128 bit flips	Corrects burst errors up to maximum theoretical bound without data loss.
Secret Recovery	Secret Byte Equality	**100.0% Exact Match**	Formal Equality	original_secret.data == recovered_secret.data strictly evaluated to TRUE.

4. Project Completion Status & Milestone Matrix

Phase-by-Phase Completion Matrix: Auditing deliverables against the PaladinMedia 6-Phase Master Architecture:

Phase ID	Milestone & Architecture Scope	Target Baseline	Completion	Delivery Status, Engineering Artifacts & Evidence
Phase 0	Sprint-0 Readiness & Specs	Governance Docs	**100%**	COMPLETE: 4 specifications baselined; manifest and dataset gates verified.
Phase 1	Cryptographic Protection & ECC	Sprint 1 Foundation	**100%**	COMPLETE: AES-256-GCM + RS GF(2^8) verified; 24/24 invariant checks passed.
Phase 2	Signal Processing & Carrier DWT	Foundation MVP 0	**100%**	COMPLETE: Haar DWT/IDWT, mask, and E03 verified (PSNR 75.71dB, 99 tests pass).
Phase 3	ACS-Sub-INN Neural Predictor	Context CNN Model	0%	PENDING: Architecture specified in Research Doc; awaiting training harness.
Phase 4	Model Training & Evaluation	800 DIV2K Images	0%	PENDING: DIV2K dataset staged in workspace; training scheduled post-Phase 3.
Phase 5	Edge Optimization & Quantization	ONNX Opset 14 / INT8	0%	PENDING: Quantization-aware training (QAT) to follow neural model convergence.
Phase 6	Laboratory GUI & Telemetry	FastAPI + Vite + Energy	**35%**	IN PROGRESS: FastAPI server & event adapter operational; Vite UI scaffolded.

Overall Completion Assessment:

• **Core Foundational Pipeline (MVP 0 — Phases 0, 1, 2): 100% COMPLETE & OPERATIONAL.** The full sender-to-receiver data path (serialization, AES-256-GCM, Reed-Solomon GF(2^8), 2D Haar DWT, mask generation, PNG storage, extraction, decryption, and 100% exact byte recovery) is fully engineered, covered by 99 automated tests, and verified against negative security fault vectors.

• **Total Multi-Phase Research Scope (Phases 0–6): ~45% COMPLETE.** Foundational cryptography, signal transforms, and backend API are complete; learned neural prediction (Phase 3), DIV2K model training (Phase 4), edge INT8 quantization (Phase 5), and CodeCarbon telemetry (Phase 6) constitute the active forward roadmap.

Observed Boundaries & Non-Claims: (1) Lossless Container Boundary: Validated exclusively over lossless PNG formats; lossy JPEG recompression is slated for Phase 3 neural robustness studies. (2) Capacity Limit: Constrained to ~7,088 B secret data per 512×512 carrier in this baseline; oversized payloads are safely rejected by hard gate. (3) Academic Stance: PaladinMedia is an experimental platform, making no claims of operational military certification or unconditional 'unbreakability'.

Next Immediate Milestones: Finalize Vite UI frontend connection to FastAPI backend; integrate CodeCarbon telemetry; initialize ACS-Sub-INN model definition for Phase 3 neural context prediction.